

Vector-Valued Functions

ANSWER KEY



Position, velocity, and acceleration vectors

- 1 A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

$$\vec{v}(t) = \langle 2t, 3t^2 \rangle. \vec{a}(t) = \langle 2, 6t \rangle.$$

- 2 For the same particle, find the speed at $t = 2$.

$$\vec{v}(2) = \langle 4, 12 \rangle. \text{Speed} = |\vec{v}(2)| = \sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}.$$

Integrating vector-valued functions

- 3 Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

$$\vec{r}(t) = \langle t^3 + C_1, t^2 + C_2 \rangle. \text{Using } \vec{r}(0) = \langle 1, 4 \rangle: C_1 = 1, C_2 = 4. \vec{r}(t) = \langle t^3 + 1, t^2 + 4 \rangle.$$

Total distance traveled

- 4 A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.

$$\text{Speed} = |\vec{v}(t)| = \sqrt{\cos^2 t + \sin^2 t} = 1 \text{ (constant)}. \text{Distance} = \int_0^\pi 1 \, dt = \pi.$$

Vectors and parametric motion combined

- 5 A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

$$\vec{v}(t) = \langle e^t, -e^{-t} \rangle. \text{At } t = 0: \vec{v}(0) = \langle 1, -1 \rangle. \text{Speed} = \sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

- 6 For the particle above, find the magnitude of the acceleration vector at $t = 0$.

$$\vec{a}(t) = \langle e^t, e^{-t} \rangle. \text{At } t = 0: \vec{a}(0) = \langle 1, 1 \rangle. |\vec{a}(0)| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Unit tangent vectors

- 7 For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

$$\vec{v}(t) = \langle 1, 2t \rangle; \text{at } t = 1: \vec{v}(1) = \langle 1, 2 \rangle, |\vec{v}(1)| = \sqrt{1 + 4} = \sqrt{5}. \vec{T}(1) = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle.$$

- 8 Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.

Dividing any nonzero vector by its own magnitude produces a vector with magnitude $\frac{|\vec{v}(t)|}{|\vec{v}(t)|} = 1$ by definition — this is the standard construction of a unit vector in the same direction.

Motion on a circle and centripetal acceleration

- 9 A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).

$\vec{v}(t) = \langle -3\sin t, 3\cos t \rangle$. $\vec{a}(t) = \langle -3\cos t, -3\sin t \rangle = -\vec{r}(t)$. Since $\vec{a}(t)$ is exactly -1 times $\vec{r}(t)$, it points directly toward the origin (centripetal acceleration).

- 10 For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t , and interpret this geometrically.

$\vec{v}(t) \cdot \vec{r}(t) = (-3\sin t)(3\cos t) + (3\cos t)(3\sin t) = -9\sin t \cos t + 9\sin t \cos t = 0$. This confirms the velocity vector is always perpendicular to the position vector — consistent with circular motion, where velocity is tangent to the circle.

Displacement vs. distance for vector motion

- 11 A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

$\vec{r}(0) = \langle 0, 0 \rangle$; $\vec{r}(2) = \langle 4, 2 \rangle$. Displacement = $\vec{r}(2) - \vec{r}(0) = \langle 4, 2 \rangle$.

- 12 Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

$\vec{v}(t) = \langle 2t, 1 \rangle$. Distance = $\int_0^2 |\vec{v}(t)| dt = \int_0^2 \sqrt{(2t)^2 + 1^2} dt = \int_0^2 \sqrt{4t^2 + 1} dt$.

Vector operations review

- 13 Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.

$2\vec{u} = \langle 6, -4 \rangle$; $3\vec{w} = \langle -3, 12 \rangle$. $2\vec{u} - 3\vec{w} = \langle 6 - (-3), -4 - 12 \rangle = \langle 9, -16 \rangle$.

- 14 Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

$\vec{u} \cdot \vec{w} = 3(-1) + (-2)(4) = -3 - 8 = -11$. Since the dot product is not 0, the vectors are NOT perpendicular.

Curvature-related speed problems

- 15 A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.

$\vec{v}(t) = \langle 3t^2, 2t \rangle$. $\text{Speed}^2 = 9t^4 + 4t^2 = 13$. Let $w = t^2$: $9w^2 + 4w - 13 = 0$. Using the quadratic formula:
 $w = \frac{-4 \pm \sqrt{16 + 468}}{18} = \frac{-4 \pm 22}{18}$. Taking the positive root: $w = 1$, so $t^2 = 1 \Rightarrow t = \pm 1$.

- 16 For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ ($t > 0$), find the velocity vector and speed at $t = 1$.

$\vec{v}(t) = \langle 1, \frac{1}{t} \rangle$. At $t = 1$: $\vec{v}(1) = \langle 1, 1 \rangle$. $\text{Speed} = \sqrt{1^2 + 1^2} = \sqrt{2}$.

Projectile motion as a vector-valued function

- 17 A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).

$\vec{v}(t) = \langle 20, 30 - 9.8t \rangle$. Maximum height occurs when $30 - 9.8t = 0 \Rightarrow t = \frac{30}{9.8} \approx 3.06$ s.

- 18 For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

$\vec{v}(0) = \langle 20, 30 \rangle$. $\text{Speed} = \sqrt{20^2 + 30^2} = \sqrt{400 + 900} = \sqrt{1300} \approx 36.1$ m/s. The horizontal component (20 m/s) stays constant throughout the flight, while the vertical component (30 m/s initially) decreases due to gravity.

Finding when a particle is at rest

- 19 A particle moves along $\vec{r}(t) = \langle t^3 - 3t, t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

$\vec{v}(t) = \langle 3t^2 - 3, 2t \rangle$. Need $3t^2 - 3 = 0$ AND $2t = 0$ simultaneously. From the second equation, $t = 0$; check the first: $3(0)^2 - 3 = -3 \neq 0$. There is NO time when both components are zero at once, so the particle is never fully at rest.

- 20 For the same particle, find the times when only the horizontal velocity component is 0.

$3t^2 - 3 = 0 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$.

Vector-valued functions and arc length

- 21 State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

$L = \int_a^b |\vec{v}(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ — arc length is the integral of speed over time, which matches the intuition that distance traveled equals the accumulated speed.

- 22 Find the arc length of $\vec{r}(t) = \langle 4t, 3t \rangle$ for $t \in [0, 5]$ using the formula above.

$\vec{v}(t) = \langle 4, 3 \rangle$, so $|\vec{v}(t)| = \sqrt{16 + 9} = 5$ (constant speed). $L = \int_0^5 5 \, dt = 25$.

Multiple choice — easy 15

- 23 Given $\vec{u} = \langle 7, -1 \rangle$ and $\vec{w} = \langle 3, 8 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 23, 22 \rangle$ ✓

b. $\langle 24, 22 \rangle$

c. $\langle 27, 13 \rangle$

d. $\langle 23, -22 \rangle$

$2\vec{u} = \langle 14, -2 \rangle$; $3\vec{w} = \langle 9, 24 \rangle$. Sum: $\langle 23, 22 \rangle$.

- 24 Given $\vec{u} = \langle 6, -8 \rangle$ and $\vec{w} = \langle -1, 6 \rangle$, find $3\vec{u} - 2\vec{w}$.

a. $\langle -15, 34 \rangle$

b. $\langle 20, 36 \rangle$

c. $\langle 21, -36 \rangle$

d. $\langle 20, -36 \rangle$ ✓

$3\vec{u} = \langle 18, -24 \rangle$; $-2\vec{w} = \langle 2, -12 \rangle$. Sum: $\langle 20, -36 \rangle$.

- 25 Given $\vec{u} = \langle 7, 7 \rangle$ and $\vec{w} = \langle 2, 8 \rangle$, find $-2\vec{u} + 3\vec{w}$.

a. $\langle -7, 10 \rangle$

b. $\langle 17, 5 \rangle$

c. $\langle -8, -10 \rangle$

d. $\langle -8, 10 \rangle$ ✓

$-2\vec{u} = \langle -14, -14 \rangle$; $3\vec{w} = \langle 6, 24 \rangle$. Sum: $\langle -8, 10 \rangle$.

26 Given $\vec{u} = \langle -5, 0 \rangle$ and $\vec{w} = \langle -8, -4 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle 14, 12 \rangle$ ✓

b. $\langle 15, 12 \rangle$

c. $\langle -1, -8 \rangle$

d. $\langle 14, -12 \rangle$

$2\vec{u} = \langle -10, 0 \rangle$; $-3\vec{w} = \langle 24, 12 \rangle$. Sum: $\langle 14, 12 \rangle$.

27 Given $\vec{u} = \langle 7, 3 \rangle$ and $\vec{w} = \langle 0, 6 \rangle$, find $2\vec{u} - 2\vec{w}$.

a. $\langle 14, -6 \rangle$ ✓

b. $\langle 14, 6 \rangle$

c. $\langle 15, -6 \rangle$

d. $\langle -14, 6 \rangle$

$2\vec{u} = \langle 14, 6 \rangle$; $-2\vec{w} = \langle 0, -12 \rangle$. Sum: $\langle 14, -6 \rangle$.

28 Given $\vec{u} = \langle -1, 3 \rangle$ and $\vec{w} = \langle 2, 3 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle -7, -3 \rangle$

b. $\langle -8, 3 \rangle$

c. $\langle 7, -3 \rangle$

d. $\langle -8, -3 \rangle$ ✓

$2\vec{u} = \langle -2, 6 \rangle$; $-3\vec{w} = \langle -6, -9 \rangle$. Sum: $\langle -8, -3 \rangle$.

29 Given $\vec{u} = \langle -7, -2 \rangle$ and $\vec{w} = \langle 8, -3 \rangle$, find $3\vec{u} - 2\vec{w}$.

a. $\langle -37, 0 \rangle$ ✓

b. $\langle -39, 1 \rangle$

c. $\langle -36, 0 \rangle$

d. $\langle 38, -5 \rangle$

$3\vec{u} = \langle -21, -6 \rangle$; $-2\vec{w} = \langle -16, 6 \rangle$. Sum: $\langle -37, 0 \rangle$.

30 Given $\vec{u} = \langle 7, -8 \rangle$ and $\vec{w} = \langle -4, 8 \rangle$, find $-2\vec{u} + 3\vec{w}$.

a. $\langle 29, -40 \rangle$

b. $\langle -25, 40 \rangle$

c. $\langle -26, -40 \rangle$

d. $\langle -26, 40 \rangle$ ✓

$-2\vec{u} = \langle -14, 16 \rangle$; $3\vec{w} = \langle -12, 24 \rangle$. Sum: $\langle -26, 40 \rangle$.

31 Given $\vec{u} = \langle -5, 1 \rangle$ and $\vec{w} = \langle 5, -4 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle -5, -5 \rangle$

b. $\langle 5, 10 \rangle$

c. $\langle 5, -10 \rangle$ ✓

d. $\langle 6, -10 \rangle$

$2\vec{u} = \langle -10, 2 \rangle$; $3\vec{w} = \langle 15, -12 \rangle$. Sum: $\langle 5, -10 \rangle$.

32 Given $\vec{u} = \langle 6, -7 \rangle$ and $\vec{w} = \langle -7, -7 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle -9, 35 \rangle$

b. $\langle -9, -35 \rangle$ ✓

c. $\langle -8, -35 \rangle$

d. $\langle 4, -35 \rangle$

$2\vec{u} = \langle 12, -14 \rangle$; $3\vec{w} = \langle -21, -21 \rangle$. Sum: $\langle -9, -35 \rangle$.

33 Given $\vec{u} = \langle 2, 0 \rangle$ and $\vec{w} = \langle 7, -1 \rangle$, find $-2\vec{u} + 3\vec{w}$.

a. $\langle 17, -3 \rangle$ ✓

b. $\langle 18, -3 \rangle$

c. $\langle 17, 3 \rangle$

d. $\langle -8, 2 \rangle$

$-2\vec{u} = \langle -4, 0 \rangle$; $3\vec{w} = \langle 21, -3 \rangle$. Sum: $\langle 17, -3 \rangle$.

34 Given $\vec{u} = \langle 1, 3 \rangle$ and $\vec{w} = \langle 5, 4 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 17, -18 \rangle$

b. $\langle 13, 17 \rangle$

c. $\langle 18, 18 \rangle$

d. $\langle 17, 18 \rangle$ ✓

$2\vec{u} = \langle 2, 6 \rangle$; $3\vec{w} = \langle 15, 12 \rangle$. Sum: $\langle 17, 18 \rangle$.

35 Given $\vec{u} = \langle 8, -8 \rangle$ and $\vec{w} = \langle -2, -4 \rangle$, find $3\vec{u} + 2\vec{w}$.

a. $\langle 10, -28 \rangle$

b. $\langle 21, -32 \rangle$

c. $\langle 20, 32 \rangle$

d. $\langle 20, -32 \rangle$ ✓

$3\vec{u} = \langle 24, -24 \rangle$; $2\vec{w} = \langle -4, -8 \rangle$. Sum: $\langle 20, -32 \rangle$.

36 Given $\vec{u} = \langle 8, 0 \rangle$ and $\vec{w} = \langle 3, -1 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle -18, -2 \rangle$

b. $\langle 7, -3 \rangle$

c. $\langle 7, 3 \rangle$ ✓

d. $\langle 8, 3 \rangle$

$2\vec{u} = \langle 16, 0 \rangle$; $-3\vec{w} = \langle -9, 3 \rangle$. Sum: $\langle 7, 3 \rangle$.

37 Given $\vec{u} = \langle -6, 8 \rangle$ and $\vec{w} = \langle 7, 5 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 10, 31 \rangle$

b. $\langle 9, 31 \rangle$ ✓

c. $\langle 9, -31 \rangle$

d. $\langle -4, 34 \rangle$

$2\vec{u} = \langle -12, 16 \rangle$; $3\vec{w} = \langle 21, 15 \rangle$. Sum: $\langle 9, 31 \rangle$.

Multiple choice — medium 15

38 A particle moves along $\vec{r}(t) = \langle 5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 26 ✓

b. 28

c. 34

d. 13

$\vec{a}(t) = \langle 10, 24 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{10^2 + 24^2} = \sqrt{676} = 26$.

39 A particle moves along $\vec{r}(t) = \langle 12t^2, -35t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 94

b. 37

c. 74 ✓

d. 76

$\vec{a}(t) = \langle 24, -70 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{24^2 + (-70)^2} = \sqrt{5476} = 74$.

40 A particle moves along $\vec{r}(t) = \langle 8t^2, 15t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 36

b. 17

c. 34 ✓

d. 46

$$\vec{a}(t) = \langle 16, 30 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{16^2 + 30^2} = \sqrt{1156} = 34.$$

41 A particle moves along $\vec{r}(t) = \langle -8t^2, 15t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 36

b. 34 ✓

c. 46

d. 17

$$\vec{a}(t) = \langle -16, 30 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-16)^2 + 30^2} = \sqrt{1156} = 34.$$

42 A particle moves along $\vec{r}(t) = \langle -4t^2, 3t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 10 ✓

b. 14

c. 12

d. 5

$$\vec{a}(t) = \langle -8, 6 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-8)^2 + 6^2} = \sqrt{100} = 10.$$

- 43 A particle moves along $\vec{r}(t) = \langle -24t^2, 7t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

- a. 62
- b. 50 ✓
- c. 52
- d. 25

$$\vec{a}(t) = \langle -48, 14 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-48)^2 + 14^2} = \sqrt{2500} = 50.$$

- 44 A particle moves along $\vec{r}(t) = \langle 35t^2, -12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

- a. 37
- b. 74 ✓
- c. 76
- d. 94

$$\vec{a}(t) = \langle 70, -24 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{70^2 + (-24)^2} = \sqrt{5476} = 74.$$

- 45 A particle moves along $\vec{r}(t) = \langle -9t^2, -40t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

- a. 98
- b. 84
- c. 82 ✓
- d. 41

$$\vec{a}(t) = \langle -18, -80 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-18)^2 + (-80)^2} = \sqrt{6724} = 82.$$

46 A particle moves along $\vec{r}(t) = \langle 24t^2, 7t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

- a. 62
- b. 52
- c. 50 ✓
- d. 25

$$\vec{a}(t) = \langle 48, 14 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{48^2 + 14^2} = \sqrt{2500} = 50.$$

47 A particle moves along $\vec{r}(t) = \langle -8t^2, -6t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

- a. 28
- b. 22
- c. 20 ✓
- d. 10

$$\vec{a}(t) = \langle -16, -12 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-16)^2 + (-12)^2} = \sqrt{400} = 20.$$

48 A particle moves along $\vec{r}(t) = \langle 6t^2, -8t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

- a. 22
- b. 20 ✓
- c. 28
- d. 10

$$\vec{a}(t) = \langle 12, -16 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{12^2 + (-16)^2} = \sqrt{400} = 20.$$

49 A particle moves along $\vec{r}(t) = \langle 6t^2, 8t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 20 ✓

b. 10

c. 28

d. 22

$$\vec{a}(t) = \langle 12, 16 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{12^2 + 16^2} = \sqrt{400} = 20.$$

50 A particle moves along $\vec{r}(t) = \langle 7t^2, 24t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 62

b. 25

c. 52

d. 50 ✓

$$\vec{a}(t) = \langle 14, 48 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{14^2 + 48^2} = \sqrt{2500} = 50.$$

51 A particle moves along $\vec{r}(t) = \langle -8t^2, 6t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 20 ✓

b. 10

c. 28

d. 22

$$\vec{a}(t) = \langle -16, 12 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-16)^2 + 12^2} = \sqrt{400} = 20.$$

52 A particle moves along $\vec{r}(t) = \langle -5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 13

b. 26 ✓

c. 34

d. 28

$$\vec{a}(t) = \langle -10, 24 \rangle \text{ (constant). } |\vec{a}(t)| = \sqrt{(-10)^2 + 24^2} = \sqrt{676} = 26.$$

Multiple choice — hard 15

53 A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. -2

b. 6

c. 2 ✓

d. 3

$$\vec{v}(t) = \langle 3t^2 - 12, 2t \rangle. \text{ Setting } 3t^2 - 12 = 0: t^2 = 4 \Rightarrow t = \pm 2. \text{ The positive value is } 2.$$

54 A particle moves along $\vec{r}(t) = \langle t^3 - 108t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 6 ✓

b. 18

c. 7

d. -6

$$\vec{v}(t) = \langle 3t^2 - 108, 2t \rangle. \text{ Setting } 3t^2 - 108 = 0: t^2 = 36 \Rightarrow t = \pm 6. \text{ The positive value is } 6.$$

- 55 A particle moves along $\vec{r}(t) = \langle t^3 - 48t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 4 ✓

b. 5

c. 12

d. -4

$\vec{v}(t) = \langle 3t^2 - 48, 2t \rangle$. Setting $3t^2 - 48 = 0$: $t^2 = 16 \Rightarrow t = \pm 4$. The positive value is 4.

- 56 A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 - 6 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 2 ✓

b. 3

c. 6

d. -2

$\vec{v}(t) = \langle 3t^2 - 12, 2t \rangle$. Setting $3t^2 - 12 = 0$: $t^2 = 4 \Rightarrow t = \pm 2$. The positive value is 2.

- 57 A particle moves along $\vec{r}(t) = \langle t^3 - 75t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 5 ✓

b. 6

c. -5

d. 15

$\vec{v}(t) = \langle 3t^2 - 75, 2t \rangle$. Setting $3t^2 - 75 = 0$: $t^2 = 25 \Rightarrow t = \pm 5$. The positive value is 5.

- 58 A particle moves along $\vec{r}(t) = \langle 7t, 48t - 4t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 7

b. 6 ✓

c. 12

d. 5

$\vec{v}(t) = \langle 7, 48 - 8t \rangle$. Setting $48 - 8t = 0$: $t = \frac{-(48)}{-8} = 6$.

- 59 A particle moves along $\vec{r}(t) = \langle 4t, 6t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 1 ✓

b. 2

c. 3

d. 4

$\vec{v}(t) = \langle 4, 6 - 6t \rangle$. Setting $6 - 6t = 0$: $t = \frac{-(6)}{-6} = 1$.

- 60 A particle moves along $\vec{r}(t) = \langle 4t, 20t - 2t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 10

b. 6

c. 5 ✓

d. 4

$\vec{v}(t) = \langle 4, 20 - 4t \rangle$. Setting $20 - 4t = 0$: $t = \frac{-(20)}{-4} = 5$.

- 61 A particle moves along $\vec{r}(t) = \langle 6t, 36t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

- a. 12
- b. 7
- c. 5
- d. 6 ✓

$\vec{v}(t) = \langle 6, 36 - 6t \rangle$. Setting $36 - 6t = 0$: $t = \frac{-(36)}{-6} = 6$.

- 62 A particle moves along $\vec{r}(t) = \langle -6t, 36t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

- a. 7
- b. 12
- c. 5
- d. 6 ✓

$\vec{v}(t) = \langle -6, 36 - 6t \rangle$. Setting $36 - 6t = 0$: $t = \frac{-(36)}{-6} = 6$.

- 63 Find the arc length of $\vec{r}(t) = \langle 9t, 40t \rangle$ for $t \in [0, 2]$.

- a. 82 ✓
- b. 98
- c. 123
- d. 84

$\vec{v}(t) = \langle 9, 40 \rangle$, so $|\vec{v}(t)| = \sqrt{9^2 + 40^2} = 41$ (constant speed). $L = \int_0^2 41 \, dt = 41(2) = 82$.

64 Find the arc length of $\vec{r}(t) = \langle 8t, 15t \rangle$ for $t \in [0, 7]$.

a. 119 ✓

b. 136

c. 161

d. 126

$$\vec{v}(t) = \langle 8, 15 \rangle, \text{ so } |\vec{v}(t)| = \sqrt{8^2 + 15^2} = 17 \text{ (constant speed)}. L = \int_0^7 17 \, dt = 17(7) = 119.$$

65 Find the arc length of $\vec{r}(t) = \langle 20t, 21t \rangle$ for $t \in [0, 4]$.

a. 164

b. 145

c. 120

d. 116 ✓

$$\vec{v}(t) = \langle 20, 21 \rangle, \text{ so } |\vec{v}(t)| = \sqrt{20^2 + 21^2} = 29 \text{ (constant speed)}. L = \int_0^4 29 \, dt = 29(4) = 116.$$

66 Find the arc length of $\vec{r}(t) = \langle 9t, 40t \rangle$ for $t \in [0, 4]$.

a. 164 ✓

b. 205

c. 196

d. 168

$$\vec{v}(t) = \langle 9, 40 \rangle, \text{ so } |\vec{v}(t)| = \sqrt{9^2 + 40^2} = 41 \text{ (constant speed)}. L = \int_0^4 41 \, dt = 41(4) = 164.$$

67 Find the arc length of $\vec{r}(t) = \langle 8t, 15t \rangle$ for $t \in [0, 3]$.

a. 68

b. 54

c. 69

d. 51 ✓

$$\vec{v}(t) = \langle 8, 15 \rangle, \text{ so } |\vec{v}(t)| = \sqrt{8^2 + 15^2} = 17 \text{ (constant speed)}. L = \int_0^3 17 \, dt = 17(3) = 51.$$